

# Critical path

When slack is bad, or even when it is good, you still need the path

# Goldens to remember

- Critical path has six pins
- Path delay equals  $A(\text{out})=3.2$
- Always match  $A(u)+d$  to  $A(v)$
- Keep these numbers handy, the browser challenges and Track A tests use the same instance

# Pseudocode

- Critical path pseudocode walks backward from the sink
- Breaking ties toward larger arrival
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

# Algorithm sketch

- The golden path is six pins from in through both cells to out
- Path delay must equal arrival at out, three point two on this instance

## Algorithm sketch — try these

```
INPUT: G, A[], sink
OUTPUT: pin path (source→...→sink)
path←[sink]; cur←sink
while cur has preds:
    pick u→cur with A[u]+delay == A[cur]
    (tie-break: largest A[u])
    prepend u; cur←u
return path
GOLDEN: in→u1/A→u1/Y→u2/A→u2/Y→out
```

# Start from the worst endpoint

## CRITICAL PATH

### Start from the worst endpoint

Step 1 / 5 · worst-sink · module02-03-critical-path



#### Explanation

Critical-path traceback begins at the endpoint with the worst setup slack. Here that is out with slack 6.8—still the only sink.

- Pick the worst slack sink
- Trace into arrival tags
- Do not guess from the source

```
sink: out
setup slack: 6.8
```

Start from the worst endpoint

# Step to the matching predecessor

CRITICAL PATH

## Step to the matching predecessor

Step 2 / 5 · match-pred · module02-03-critical-path



### Explanation

At each pin, choose a predecessor  $u$  where  $A(u) + \text{delay}$  equals  $A(v)$ . That arc is on the critical path.

- Match  $A(u) + d$  to  $A(v)$
- Ties: prefer larger  $A(u)$
- Walk until a source

```
out ← u2/Y (+0.2)
u2/Y ← u2/A (+1.5)
```

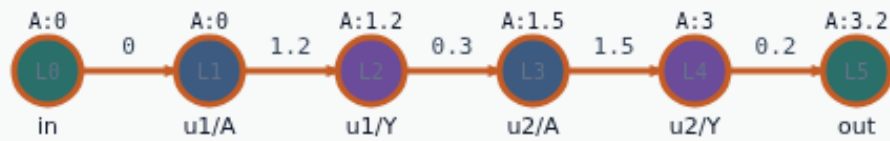
Step to the matching predecessor

# The full golden path

CRITICAL PATH

## The full golden path

Step 3 / 5 · full-path · module02-03-critical-path



### Explanation

The complete critical path is  $\text{in} \rightarrow \text{u1/A} \rightarrow \text{u1/Y} \rightarrow \text{u2/A} \rightarrow \text{u2/Y} \rightarrow \text{out}$ . Arrival at out equals the sum of arc delays along this path: 3.2.

- Six pins on one path
- Path delay = A(out)
- Golden list matches PROP\_GOLDENS

$\text{in} \rightarrow \text{u1/A} \rightarrow \text{u1/Y} \rightarrow \text{u2/A} \rightarrow \text{u2/Y} \rightarrow \text{out}$   
path delay = 3.2

The full golden path

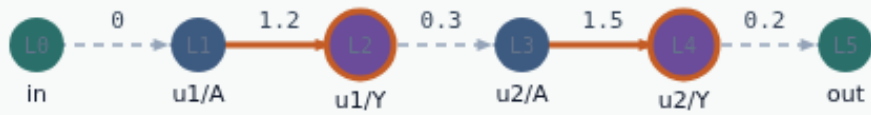


# Fix the path, not a random gate

## CRITICAL PATH

### Fix the path, not a random gate

Step 4 / 5 · why-it-matters · module02-03-critical-path



#### Explanation

Optimization and ECO work chase critical paths. Tag matching keeps you honest when reconvergence exists—on this chain there is only one route.

- Trace before you resize
- Reconvergence needs care
- Toy chain has one route

reconvergence: none  
cells on path: u1, u2

Fix the path, not a random gate

# Slack and path travel together

CRITICAL PATH

## Slack and path travel together

Step 5 / 5 · slack-path · module02-03-critical-path



### Explanation

Reports pair worst slack with its path. Next labs edit delays and exceptions—always re-trace after the tags change.

- Slack names the problem
- Path names where to look
- Re-trace after edits

```
slack 6.8 @ out  
path length 6
```

Slack and path travel together

# Browser lab track

- In the browser lab, open **critical-path**
- Load the starter, run the analysis once, and read the metrics panel
- Orient yourself, challenge panel, canvas, Check, then mirror the same goldens in code

# Implement track

- In the implement track
- Run ``python3 common/test_propagate.py`` (and the timing-graph test) until the goldens print

# Pitfall

- Do not mix setup and hold required maps
- Do not propagate before the graph is levelized
- After an edit or exception, recompute, stale tags lie

## Your turn

- Finish the checklist on at least one track, preferably both
- When your numbers match the goldens, take the quiz, then continue